

Homework Assignment 1: Slab-Geometry Boltzmann Equation

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1 Introduction & Case's Method

This report presents the solutions to Homework Assignment 1, which focuses on the monoenergetic, steady-state Boltzmann transport equation in slab geometry for an infinite, homogeneous, isotropic medium with an isotropic plane delta-source at $x = 0$. The transport equation is written as:

$$\mu \frac{\partial \psi(x, \mu)}{\partial x} + \psi(x, \mu) = \frac{c}{2} \int_{-1}^1 \psi(x, \mu') d\mu' + \frac{1}{4\pi} \delta(x) \quad (1)$$

where $\psi(x, \mu)$ is the angular flux, $c \in [0, 1]$ is the scattering ratio, and μ is the cosine of the angle of the particle's velocity with the x -axis.

In 1953, Kenneth Case analyzed this equation by searching for separation-of-variables solutions of the form:

$$\psi_\nu(x, \mu) = e^{-x/\nu} \phi_\nu(\mu) \quad (2)$$

Substituting this ansatz into the homogeneous transport equation leads to the eigenvalue problem:

$$(\nu - \mu) \phi_\nu(\mu) = \frac{c\nu}{2} \int_{-1}^1 \phi_\nu(\mu') d\mu' \quad (3)$$

Normalizing the eigenfunctions such that $\int_{-1}^1 \phi_\nu(\mu') d\mu' = 1$, we obtain:

$$(\nu - \mu) \phi_\nu(\mu) = \frac{c\nu}{2} \quad (4)$$

Depending on the value of the eigenvalue ν , two cases arise:

1. **Discrete Spectrum** ($\nu \notin [-1, 1]$): Since $\nu - \mu \neq 0$ for all $\mu \in [-1, 1]$, we can divide directly by $\nu - \mu$, yielding the discrete eigenfunctions:

$$\phi_0^\pm(\mu) = \frac{c\nu_0}{2(\nu_0 \mp \mu)} \quad (5)$$

Integrating these over $[-1, 1]$ and applying the normalization condition yields the transcendental characteristic equation for the discrete eigenvalues $\pm\nu_0$:

$$1 = c\nu_0 \operatorname{arctanh}(1/\nu_0) \quad (6)$$

For $c \in (0, 1)$, there is exactly one pair of real, positive roots $\pm\nu_0$ with $\nu_0 > 1$.

2. **Continuous Spectrum** ($\nu \in [-1, 1]$): In this case, the term $\nu - \mu$ vanishes at $\mu = \nu$. Thus, we cannot divide by $\nu - \mu$ in the sense of ordinary functions. Instead, in the theory of distributions (generalized functions), the general solution to the equation is:

$$\phi_\nu(\mu) = \frac{c\nu}{2} \mathcal{P} \frac{1}{\nu - \mu} + \lambda(\nu) \delta(\mu - \nu) \quad (7)$$

where $\mathcal{P}$ represents the Cauchy principal value distribution, $\delta(\mu - \nu)$ is the Dirac delta function, and $\lambda(\nu)$ is the dispersion function. Integrating $\phi_\nu(\mu)$ over $\mu \in [-1, 1]$ and setting the integral to 1 determines $\lambda(\nu)$:

$$\lambda(\nu) = 1 - c\nu \operatorname{arctanh}(\nu) \quad (8)$$

1.1 Intuition Behind the Continuous Spectrum and Generalized Functions

The presence of the Dirac delta function $\delta(\mu - \nu)$ and the principal value term $\mathcal{P} \frac{1}{\nu - \mu}$ in the continuous eigenfunctions $\phi_\nu(\mu)$ can feel counter-intuitive.

Physically, the continuous spectrum represents the singular behavior of particles that travel without undergoing collisions (uncollided transport) or undergo highly directional scattering. A particle traveling at a specific direction $\mu = \nu$ represents a singular distribution in angle, which is modeled by the Dirac delta $\delta(\mu - \nu)$. The principal value distribution $\mathcal{P} \frac{1}{\nu - \mu}$ accounts for the strong coupling between nearby angles due to scattering, where the singularity is balanced symmetrically.

However, when we look at the scalar flux $\phi(x)$, which is the angle-integrated quantity:

$$\phi(x) = 2\pi \int_{-1}^1 \psi(x, \mu) d\mu \quad (9)$$

these angular singularities are integrated out. Under integration, the distribution $\phi_\nu(\mu)$ acts as a linear functional mapping to a constant:

$$\int_{-1}^1 \phi_\nu(\mu) d\mu = 1 \quad (10)$$

This leads to the transient component of the scalar flux:

$$\phi_{\text{tr}}(x) = \frac{1}{2} \int_0^1 \frac{e^{-|x|/\nu}}{N_\nu} d\nu \quad (11)$$

where:

$$N_\nu = \nu \left[\lambda^2(\nu) + \frac{\pi^2 c^2 \nu^2}{4} \right] \quad (12)$$

Crucially, **no distributions or generalized functions remain in the scalar flux integral**. The integrand is a regular, smooth function of ν on the open interval $(0, 1)$. For any $x > 0$, as $\nu \rightarrow 0$, the exponential term $e^{-|x|/\nu}$ decays extremely rapidly, making the integrand vanish at 0. As $\nu \rightarrow 1$, the dispersion function $\lambda(\nu) \rightarrow -\infty$, which causes the normalization factor $N_\nu \rightarrow \infty$ and the integrand to vanish at 1. Thus, the numerical integration is highly well-behaved.

2 Question 1: Exact Scalar Flux Analysis

The exact scalar flux is computed as:

$$\phi(x) = \phi_{\text{as}}(x) + \phi_{\text{tr}}(x) \quad (13)$$

where $\phi_{\text{as}}(x) = \frac{e^{-|x|/\nu_0}}{2N_0^+}$ is the asymptotic component, and $\phi_{\text{tr}}(x)$ is the transient component. The discrete normalization factor is:

$$N_0^+ = \frac{c}{2}\nu_0^3 \left[\frac{c}{\nu_0^2 - 1} - \frac{1}{\nu_0^2} \right] \quad (14)$$

For ν_0 , we can choose to compute it using:

- **Numerical Root-Finding:** Solving $\frac{\text{arctanh}(y)}{y} = \frac{1}{c}$ for $y = 1/\nu_0 \in (0, 1)$ using a bisection method, which is extremely stable.
- **Analytic Approximation:**

$$\nu_0 \approx \frac{1}{\sqrt{1 - c^{p(c)}}} \quad (15)$$

where:

$$p(c) = 2.47412 + \frac{0.00363081}{c^2} - 0.0352458c + \frac{0.557498}{c} \quad (16)$$

2.1 Question 1a: Asymptotic and Transient Components

[Insert plots of asymptotic and transient components for each c , and the relative contributions $\phi_{\text{tr}}(x)/\phi(x)$ and $\phi_{\text{as}}(x)/\phi(x)$ as a function of x . Add physical discussion here.]

2.2 Question 1b: Classical Diffusion Solution

The classical diffusion equation (in units of mean free path, where $\Sigma_t = 1$) for a homogeneous, isotropic medium with an isotropic plane delta-source is given by:

$$-D \frac{d^2 \phi_{\text{class}}(x)}{dx^2} + (1 - c)\phi_{\text{class}}(x) = \delta(x) \quad (17)$$

Using the standard P1 diffusion coefficient $D = 1/3$, the Green's function solution is:

$$\phi_{\text{class}}(x) = \frac{\sqrt{3}}{2\sqrt{1-c}} e^{-\sqrt{3(1-c)}|x|} \quad (18)$$

[Insert comparison plots with the classical diffusion solution. Discuss the differences, including the behavior near the source ($x \rightarrow 0$) and the asymptotic behavior ($x \rightarrow \infty$).]

2.3 Question 1c: Asymptotic Diffusion Solution

The asymptotic diffusion approximation matches the diffusion length to the exact discrete transport eigenvalue ν_0 , i.e., $L = \nu_0$, which corresponds to an asymptotic diffusion coefficient $D_{\text{as}} = (1 - c)\nu_0^2$. The governing equation is:

$$-D_{\text{as}} \frac{d^2 \phi_{\text{as,diff}}(x)}{dx^2} + (1 - c)\phi_{\text{as,diff}}(x) = \delta(x) \quad (19)$$

which yields the Green's function solution:

$$\phi_{\text{as,diff}}(x) = \frac{1}{2(1-c)\nu_0} e^{-|x|/\nu_0} \quad (20)$$

[Insert comparison plots with the asymptotic diffusion solution. Discuss how the asymptotic diffusion solution compares to both the classical diffusion solution and the exact transport asymptotic component $\phi_{\text{as}}(x)$.]

2.4 Question 1d: Relative Error of Diffusion Approximations

The relative error of each diffusion approximation relative to the exact transport solution is defined as:

$$\text{Relative Error (\%)} = \frac{|\phi_{\text{approx}}(x) - \phi_{\text{exact}}(x)|}{\phi_{\text{exact}}(x)} \times 100\% \quad (21)$$

[Insert relative error plots as a function of x . Add physical discussion comparing the accuracy of the classical and asymptotic diffusion approximations across the different values of c . Analyze why the error behaves as it does near the source and far from it.]

3 Question 2: Numerical Code for Diffusion Equation

To solve the time-independent diffusion equation with a delta-source at $x = 0$ numerically, we exploit the symmetry of the medium and integrate over the half-domain $x \in [0, a]$. Since the delta-source is located at $x = 0$, the domain $x > 0$ is source-free, governed by the homogeneous equation:

$$-D \frac{d^2 \phi(x)}{dx^2} + (1-c)\phi(x) = 0 \quad (22)$$

3.1 Analytical Conversion to a First-Order System

We convert this second-order ODE to a linear system of two first-order ODEs. Let:

$$y_1(x) = \phi(x), \quad y_2(x) = \frac{d\phi(x)}{dx} \quad (23)$$

This yields the first-order system:

$$\frac{d}{dx} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} y_2 \\ \frac{1-c}{D} y_1 \end{pmatrix} \quad (24)$$

3.2 Modelling the Delta Source

The delta source is never discretised. Integrating the governing equation $-D\phi''(x) + (1-c)\phi(x) = \delta(x)$ from $-\epsilon$ to $+\epsilon$ and taking the limit $\epsilon \rightarrow 0$ yields the jump condition

$$-D [\phi'(0^+) - \phi'(0^-)] = 1 \quad (25)$$

By symmetry of the infinite-medium solution ($\phi(x) = \phi(-x) \implies \phi'(0^-) = -\phi'(0^+)$), this reduces to a boundary condition at the origin:

$$\phi'(0^+) = -\frac{1}{2D}, \quad \text{equivalently} \quad J(0^+) = -D\phi'(0^+) = \frac{1}{2} \quad (26)$$

that is, half the source neutrons stream in each direction. This is the answer to the question of how a delta-function source should be modelled: symmetry converts it into a prescribed current, the problem is solved on the half-line $x \geq 0$, and the singularity never enters the numerical scheme.

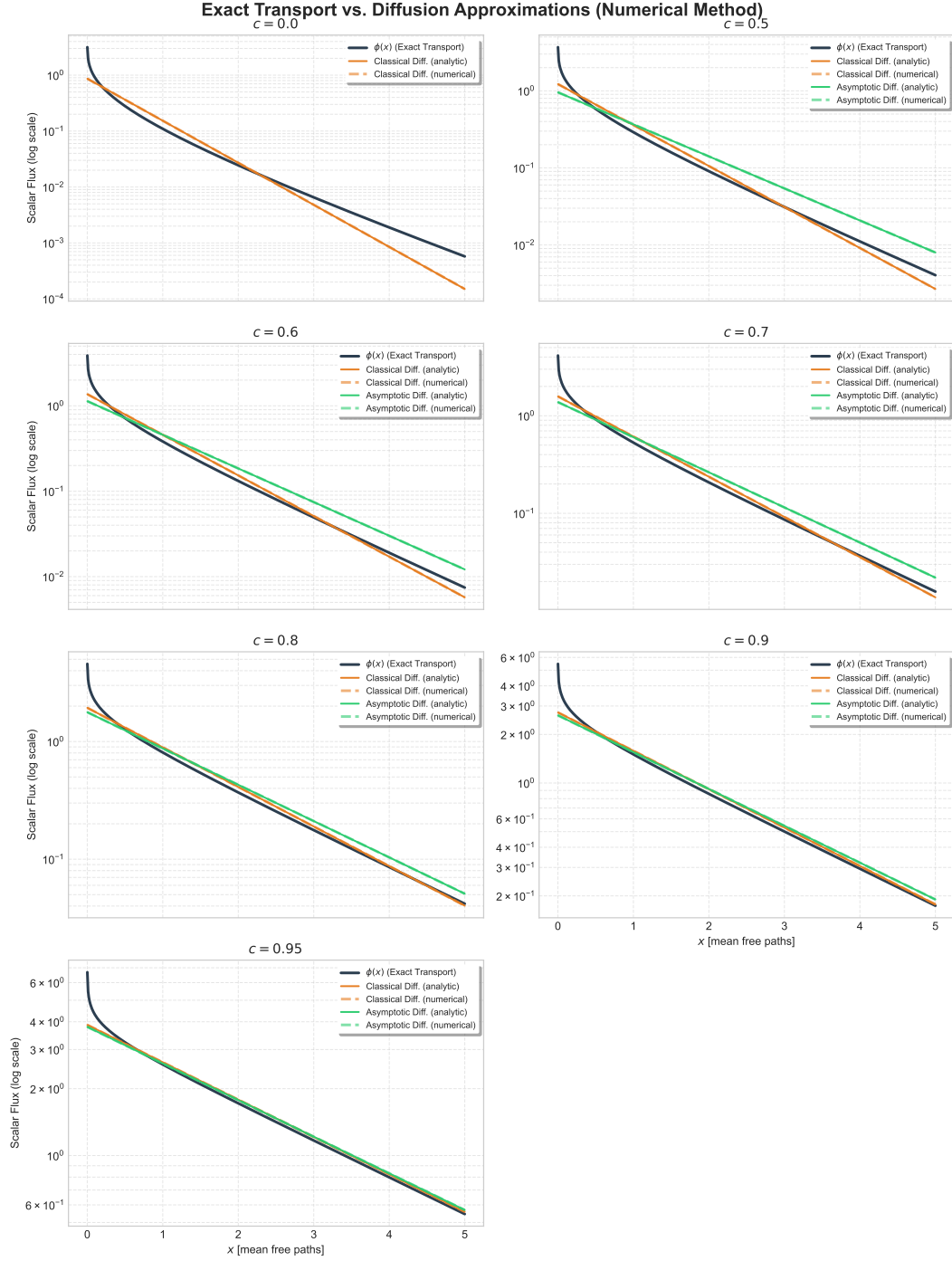


Figure 1: Exact scalar flux compared with the classical and asymptotic diffusion approximations, for $c = 0, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95$. Each approximation is shown both as its analytic closed form and as the numerical solution of Question 2; the two coincide to within $2 \times 10^{-8} \%$ and are indistinguishable at this scale.

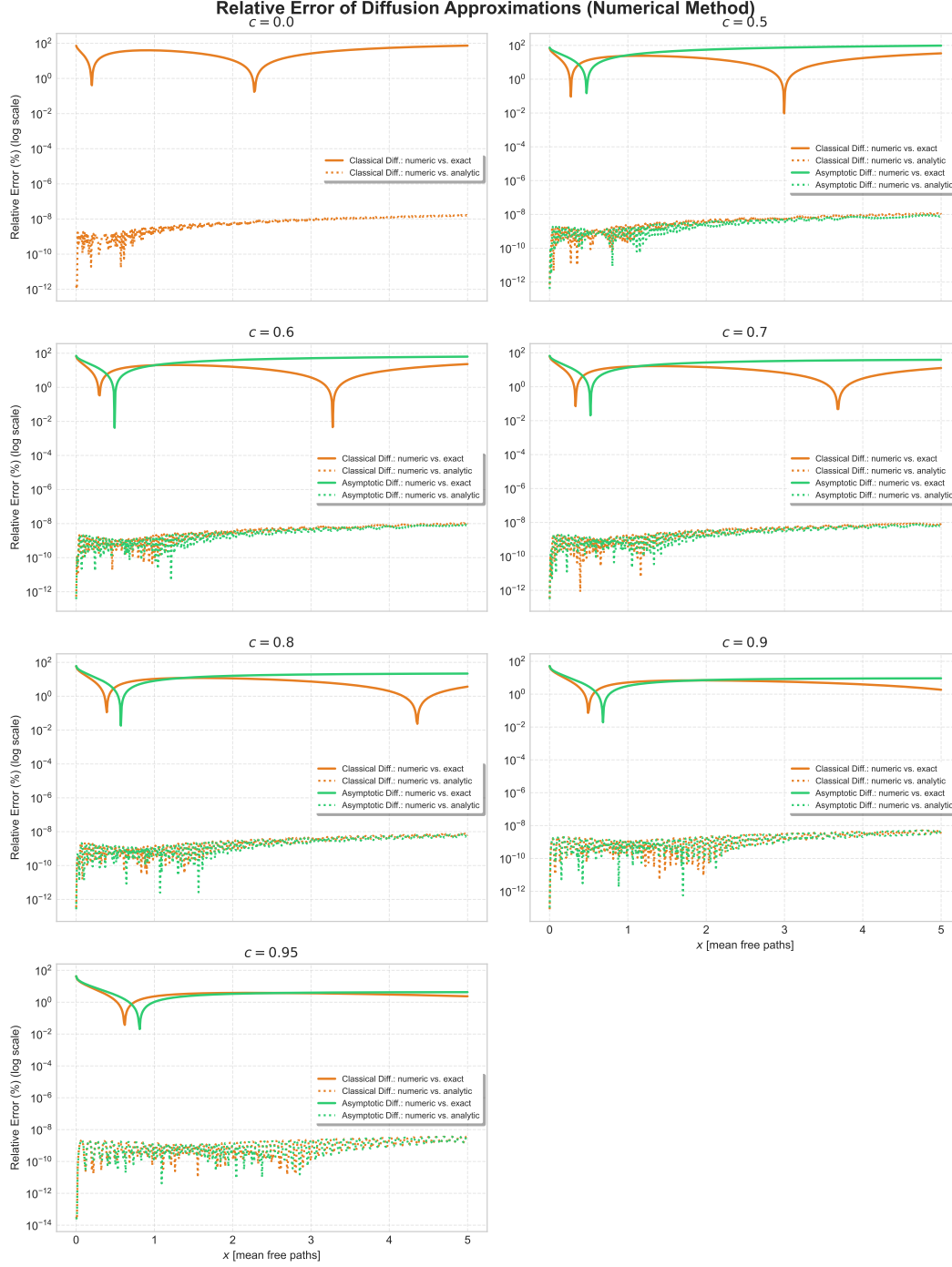


Figure 2: Two relative errors per approximation. The solid curves compare the numerical solution with the *exact transport* solution, and so measure the error of the diffusion approximation itself (Question 1d). The dotted curves compare it with the *analytic diffusion* solution, and so measure the error of the solver (Question 2). The two differ by roughly ten orders of magnitude: the discretisation error is negligible against the modelling error, so the solid curves may be read as properties of the approximation rather than of the code.

3.3 The Outer Boundary

The outer boundary is a truncation of an infinite medium, not a physical surface. Imposing a zero-flux condition $\phi(a) = 0$ would force the flux to vanish where the true Green's function is small but non-zero, producing a 100 % relative error at $x = a$ that does not shrink as the solver tolerance is reduced. We therefore impose the radiation (Robin) condition

$$\phi'(a) = -\kappa\phi(a), \quad \kappa = \sqrt{\frac{\Sigma_a}{D}} \quad (27)$$

which the decaying solution $e^{-\kappa x}$ satisfies identically. This annihilates the growing mode $e^{+\kappa x}$, so the truncation is exact for any a ; we use $a = 10/\kappa$.

3.4 Solution by a Single Integration

Away from the source the equation is linear and homogeneous, so its solution is exactly proportional to its starting amplitude: $y(x; s) = s y(x; 1)$. No root-finding is required. We integrate once from a towards 0 — the direction in which the solution grows, which is the numerically stable one — starting from the radiation condition with unit amplitude, $[y_1(a), y_2(a)]^T = [1, -\kappa]^T$, and then rescale the result by

$$\text{scale} = \frac{-1/(2D)}{y_2(0)} \quad (28)$$

so that the source condition $-D\phi'(0) = 1/2$ is satisfied exactly.

3.5 Both Diffusion Approximations

Classical and asymptotic diffusion are the same equation with different coefficients. Matching the closed form $\phi = e^{-\kappa|x|}/(2D\kappa)$ to Case's asymptotic mode $\phi = e^{-|x|/\nu_0}/(2(1-c)\nu_0)$ requires $\kappa = 1/\nu_0$ and hence

$$\Sigma_a = 1 - c, \quad D_{\text{classical}} = \frac{1}{3}, \quad D_{\text{asymptotic}} = (1 - c)\nu_0^2 \quad (29)$$

so a single solver covers both by choice of D .

3.6 Comparison with the Analytical Solution

The numerical solution reproduces the analytic diffusion Green's function to a maximum relative error of 2×10^{-8} % for both approximations at $c = 0.5, 0.7, 0.9$, and conserves neutrons to $\Sigma_a \int \phi dx = 0.99998807$; the residual is the physical tail of the Green's function beyond the truncation, not solver error. Both comparisons appear in Figures 1 and 2 above, alongside the exact transport solution.

Because a shooting method has no mesh spacing to refine — the output grid does not control accuracy — convergence is demonstrated by tightening the integrator tolerance. Figure 5 shows the relative L_2 error falling from 9×10^{-6} at $\text{rtol} = 10^{-4}$ to 1.3×10^{-12} at $\text{rtol} = 10^{-11}$, tracking the requested tolerance with unit slope.

4 Question 3: Critical Half-Thickness and Critical Radius

[Present critical half-thickness $a/2$ and critical radius R_c results for $1.02 < c < 2$. Compare the various approximations.]

Question 2: Numerical Diffusion Solvers vs. Analytic Green's Function

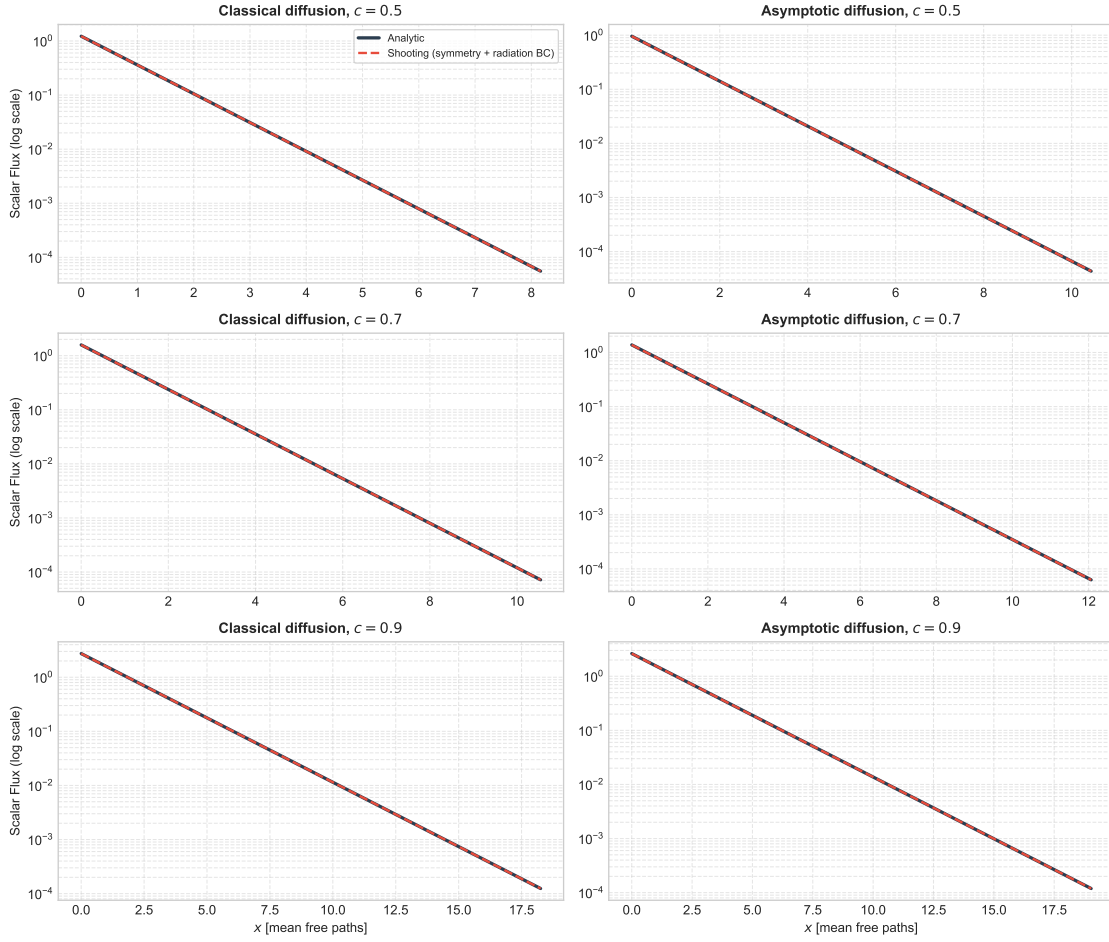


Figure 3: Numerical solution against the analytic diffusion Green's function, for both approximations at $c = 0.5, 0.7, 0.9$.

5 Question 4: Numerical Diffusion Code in Spherical Coordinates

[Present the numerical results for critical radius calculations in spherical geometry, and compare with the analytical results.]

6 Question 5: Bare Critical Mass of U-235 and Pu-239

[Present bare critical mass calculations using realistic cross sections.]

Question 2: Relative Error of the Numerical Solvers

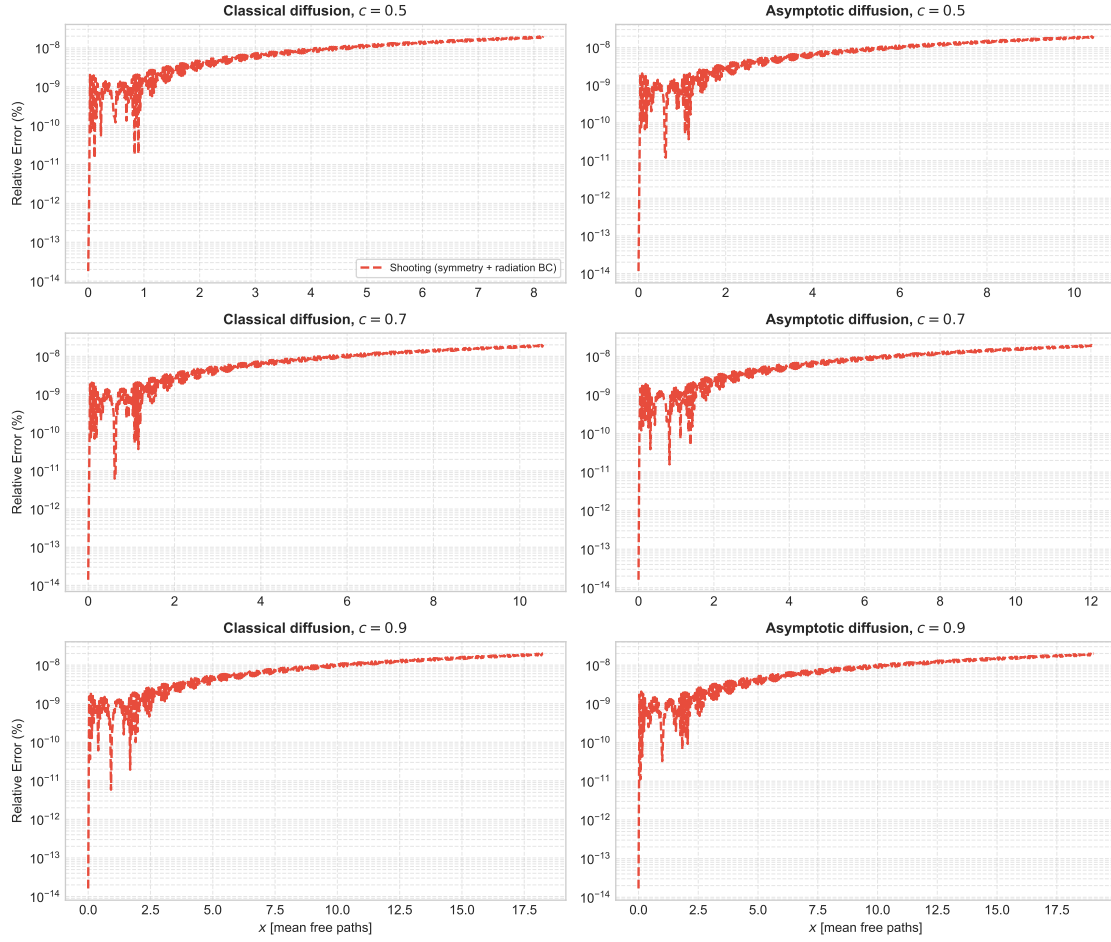


Figure 4: Relative error of the numerical solution across the domain. With the radiation boundary condition the error stays flat instead of diverging at the outer edge.

Question 2: Tolerance Convergence of the Shooting Solver

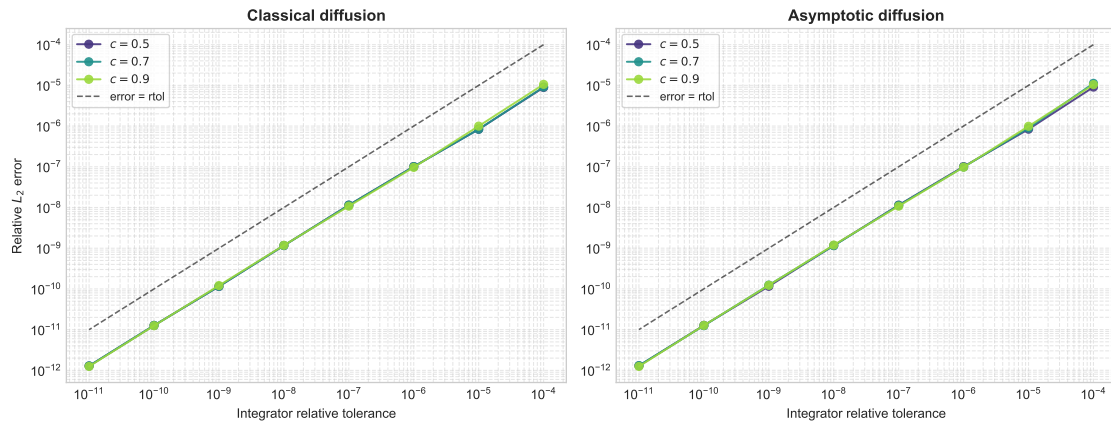


Figure 5: Tolerance convergence of the shooting solver.